

The Worker and Firm Components of Monopsony

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Abstract

Labor-market power is usually measured for the whole firm, but a firm markdown—the gap between what labor produces and what it is paid—cannot say which workers inside the firm bear it. We construct worker-level markdowns by allocating firm value added net of capital costs across occupations and workers, using production weights and human-capital-adjusted hours. In Portuguese linked employer-employee and balance-sheet data with 13.7 million worker-year observations, firm and worker differences account for large shares of markdown dispersion—75.1 and 38.8 percent after a leave-out bias correction—while their covariance is negative: high-markdown workers tend to sit at lower-markdown firms. Markdowns rise sharply up the occupation ladder, with managers and professionals marked down 93 percent more than service and manual workers. Monopsony is thus distributed unequally across workers inside the firm, not a single employer-level premium.

KEYWORDS: Monopsony, Markdowns, Worker Heterogeneity, Firm Structure

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1. Introduction

Firm markdowns are averages. They measure whether an employer pays labor less than an output-based benchmark, while hiding how that wedge is distributed inside the firm. Who bears that wedge matters. Employers combine managers, professionals, clerks, technicians, and manual workers, and these workers differ in tasks, skills, outside options, and likely labor-supply elasticities. When firms have labor-market power, the incidence of that power depends on which workers inside the firm bear the gap.

We make this within-firm incidence observable by turning a firm markdown into a worker-firm-year measure. For each worker, the measure compares earnings with the labor claim assigned to that worker inside the firm. The assignment starts from firm value added after capital costs. It allocates this amount first across broad occupation groups, then across workers within each group. The pieces add back to the firm-year labor claim, so the exercise redistributes a fixed total across workers. Under a benchmark production model, the assigned claim equals the worker’s marginal revenue product of labor; even when that benchmark is relaxed, the measure gives a transparent worker-level split of the firm’s labor claim that can be compared directly with wages in the matched panel. Our contribution is to make this worker-level wedge observable across the full linked panel and to decompose its variance into worker, firm, occupation, market, and match components.

Worker heterogeneity accounts for a large part of markdown dispersion. In our Portuguese sample, a variance decomposition of log markdowns—bias-corrected following Kline, Saggio, and Sølvssten (2020, KSS)—attributes 75.1 percent of the variance to differences across firms and 38.8 percent to differences across workers. The two components move against each other: the firm-worker covariance contributes minus 14.0 percent of total variance. High-markdown workers are partly sorted into lower-markdown employers.

We bring the measure to Portuguese administrative data linking workers to firm balance sheets. The matched employer-employee register, *Quadros de Pessoal* (QP), records workers, wages, hours, occupations, and employers. The firm balance-sheet register, *Sistema de Contas Integradas das Empresas* (SCIE), provides value added, capital, investment, depreciation, and interest expenses. The main connected-set markdown sample contains 13.7 million worker-year observations, 2.4 million workers, and 166,035 firms over 2004–2024. We first remove capital’s user cost from value added. We then allocate the remaining labor claim across occupation groups using production-function contribution weights, and within occupation groups using human-capital-adjusted hours.¹

Beyond its size in the decomposition, the firm component changes how we read wage

¹The within-cell weights use a Mincer-style index of predetermined schooling and potential experience, which keeps the worker’s own wage out of the allocated claim and avoids a mechanical link to the markdown denominator. Section 3.3 gives the calibration; a raw-hours allocation is a robustness check.

premium. We estimate same-sample firm, worker, and year decompositions for wages, allocated labor claims, and markdowns. High-wage firms are also high-claim firms: a one log point higher firm wage component is associated with a 0.57 log point higher allocated labor-claim component. Because markdowns are claims divided by wages, the negative relation between wage components and markdown components follows from the fact that claims rise by less than wages. Workers at high-wage firms therefore receive a larger share of the labor claim measured from firm accounts. This links two objects usually studied separately: firm wage premia in matched worker-firm data and production-based markdowns in firm accounts.

The occupation gradient is large. Relative to service and manual workers, technicians and clerks have conditional markdowns about 0.37 log points higher (45 percent in levels), and managers and professionals about 0.66 log points higher (93 percent in levels). The gradient reflects a gap between product and pay across occupations: the allocation gives occupation groups different production weights, while the markdown divides each group's allocated claim by its earnings. Most of the occupation gap comes from differences across firm-years. Technicians, clerks, managers, and professionals tend to work in firm-years with higher markdowns for all workers, and this sorting accounts for 79 percent of the technician/clerk gap and 70 percent of the manager/professional gap.

The market-structure gradient varies sharply across occupations. Local employment concentration predicts lower market markdown components for service and manual workers, almost no gradient for technicians and clerks, and a smaller negative gradient for managers and professionals. Within local markets, larger employers also have lower firm markdown components: a 10 percent increase in average payroll share is associated with about a 1.1 percent lower firm component. These descriptive estimates say that concentration and firm size map into worker-level wedges differently across the workforce.

Worker-firm interactions matter less than one might expect. A recent Roy-style monopsony model by Bils, Kaymak, and Wu (2025) emphasizes that firms may have the most wage-setting power over workers who are especially productive at that firm. A dominant complementarity channel would appear as a large worker-firm interaction term in markdowns. We test that concern with a grouped decomposition in the spirit of Bonhomme, Lamadon, and Manresa (2019, BLM). Allowing the markdown to differ freely across every worker-type-by-firm-class cell raises explained variance by only 1.5 percentage points over a specification with additive worker and firm terms. Most markdown dispersion is therefore captured by worker and firm components on their own.

The paper links two empirical traditions. Production-based markdown and monopsony estimates measure firm-level wage-product wedges (De Loecker and Warzynski 2012; Yeh, Macaluso, and Hershbein 2022). The linked-employer-employee literature decomposes pay into worker and firm components (Abowd, Kramarz, and Margolis 1999; Card, Heining,

and Kline 2013; Card et al. 2018). In spirit, our exercise follows variance decompositions of wage inequality: it asks how much dispersion in a worker-level outcome is associated with workers, firms, and their sorting. The outcome here is the wage-product wedge. We combine the production-side discipline of the first with the decomposition discipline of the second and apply both to worker-level markdowns.

Structural models of imperfect competition also deliver heterogeneous markdowns as estimated equilibrium objects. Berger, Herkenhoff, and Mongey (2022) estimate oligopsony with differentiated jobs. Lamadon, Mogstad, and Setzler (2022) recover firm-specific labor-supply elasticities under two-sided heterogeneity. Wong (2025) separates firm wage premia into monopoly, monopsony, and bargaining components. Closest to our setting, Jáñez and Delgado-Prieto (2025) estimate a general-equilibrium oligopsony model with firm organization on Portuguese data and find monopsony power concentrated in higher-ranked occupations. We instead measure the wedge directly from firm accounts for every worker-firm-year, rather than recovering it from an estimated equilibrium model.² The allocation trades structural completeness for transparency and panel coverage.

The incidence of labor-market power is therefore a within-firm question. A firm markdown can tell us whether an employer pays labor below an output-based benchmark. It leaves open whether the wedge is concentrated among low-wage workers, high-skill workers, particular occupations, or particular worker-firm matches. Worker-level markdowns make those margins observable. In Portugal, wage-product wedges vary across firms, workers, occupations, local markets, and, to a lesser extent, worker-firm matches.

Outline. Section 2 defines the allocated markdown and its benchmark interpretation. Section 3 describes the Portuguese data and the empirical construction. Section 4 decomposes markdown dispersion and relates the estimated components to concentration and firm size. Section 5 asks whether grouped worker-firm interactions materially improve on the additive decomposition. Section 6 concludes.

2. Framework

The framework has two jobs. First, it defines the empirical markdown measure used throughout the paper. Second, it states the benchmark assumptions under which that measure has a worker-level marginal-product and monopsony interpretation. The construction is intentionally modest: it is an accounting-consistent worker-level allocation with a narrow structural benchmark, not a full equilibrium model of assignment, amenities, bargaining, or search.

²Jáñez and Delgado-Prieto (2025), in the same Portuguese data, recover occupation-level monopsony from an estimated model of firm organization; we measure a worker-level wedge for every worker-firm-year and decompose its variance directly.

The destination is a single object: a worker-level markdown, the ratio of the labor claim a worker generates to what the worker is paid. Under the benchmarks below, this ratio equals $1 + 1/\eta_{ijt}$, the firm's wage-setting power over that worker, and it is larger the less elastic the worker's supply of labor to the firm.

2.1. The two-step allocation

This section defines the empirical measure that the rest of the framework interprets. The construction has three steps. First, remove the user cost of capital from firm-year value added to obtain the residual assigned to labor. Second, split that residual across labor groups using contribution weights from the production estimates. Third, split each group claim across workers in proportion to human-capital-adjusted hours. The result is an accounting-consistent worker-level labor claim that adds up exactly to the firm-year residual assigned to labor.

Labor-claimable value added. At time t , firm j has factor-cost value added VA_{jt} (value added net of product taxes and subsidies), capital stock K_{jt} , and workers partitioned into labor groups $g \in \mathcal{G}_{jt}$. In the empirical application, the groups are occupation groups. The allocation starts from the part of value added left after capital is paid its user cost:

$$(1) \quad VA_{jt}^L \equiv VA_{jt} - UC_{s(j)t} K_{jt}, \quad UC_{st} \equiv r_t + \delta_s,$$

where $s(j)$ denotes firm j 's sector, r_t is the common opportunity cost of capital, and δ_s is sector-specific depreciation.

VA_{jt}^L is the numerator used to construct worker-level markdowns. It removes a first-order source of variation in capital intensity before assigning the remaining firm-year value added to workers. Interpreting this residual as the total marginal-product claim of labor requires the production assumptions in Section 2.2. Without those assumptions, VA_{jt}^L is still a transparent accounting residual, but it may also contain product-market rents, fixed costs, unmeasured capital services, taxes or subsidies, and capital measurement error.

From production estimates to group weights. Let $g(i)$ denote worker i 's labor group, and let $\beta_{g,s(j)} > 0$ be the measured contribution weight of group g in sector $s(j)$. Because not every firm employs every group, the allocation normalizes these weights over the groups present in the firm-year:

$$(2) \quad \bar{\beta}_{g,s(j)} \equiv \frac{\beta_{g,s(j)}}{\sum_{h \in \mathcal{G}_{jt}} \beta_{h,s(j)}}$$

The normalized weight $\bar{\beta}_{g,s(j)}$ is the share of the labor-claimable block assigned to group g . It depends on the realized group composition $\mathcal{G}_{jt}$ through the denominator, although the notation suppresses this dependence.

Under the benchmark production technology below, $\bar{\beta}_{g,s(j)}$ is the output elasticity of group g inside the labor aggregate. Outside that benchmark, it is a maintained allocation weight. This distinction matters: the same formula defines the empirical measure in either case, but only the benchmark production assumptions turn the weight into a marginal-product share.

Within-group shares and worker-level labor claims. The second split is within group. Let h_{ijt} denote worker i 's observed annual hours at firm j in year t . Let $q_{it} > 0$ denote a calibrated efficiency-per-hour weight built from predetermined human-capital variables. The baseline uses the effective-hours index

$$(3) \quad x_{ijt} \equiv q_{it} h_{ijt}.$$

Hours are the observed input quantity. The weight q_{it} is the maintained adjustment for efficiency per hour.

Worker i 's within-group share is

$$(4) \quad \pi_{ijt} \equiv \frac{x_{ijt}}{\sum_{k \in g(i),j,t} x_{kjt}} = \frac{q_{it} h_{ijt}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}},$$

so that $\sum_{i \in g,j,t} \pi_{ijt} = 1$ for every firm-year group cell. Combining the group split and the within-group split gives worker i 's allocated labor claim:

$$(5) \quad MRPL_{ijt}^{\text{alloc}} \equiv VA_{jt}^L \cdot \bar{\beta}_{g(i),s(j)} \cdot \pi_{ijt}.$$

Substituting (2) and (4) gives

$$MRPL_{ijt}^{\text{alloc}} = VA_{jt}^L \cdot \frac{\beta_{g(i),s(j)}}{\sum_{h \in \mathcal{G}_{jt}} \beta_{h,s(j)}} \cdot \frac{q_{it} h_{ijt}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}}.$$

This formula has a useful interpretation even before imposing production structure. It assigns each worker the firm-year residual, times the worker's group share of that residual, times the worker's share within the group. Figure 1 shows the same logic for the running example with one manager and nine identical service workers.

The construction adds up exactly:

$$(6) \quad \sum_{i \in j,t} MRPL_{ijt}^{\text{alloc}} = VA_{jt}^L \sum_{g \in \mathcal{G}_{jt}} \bar{\beta}_{g,s(j)} \sum_{i \in g,j,t} \pi_{ijt} = VA_{jt}^L.$$

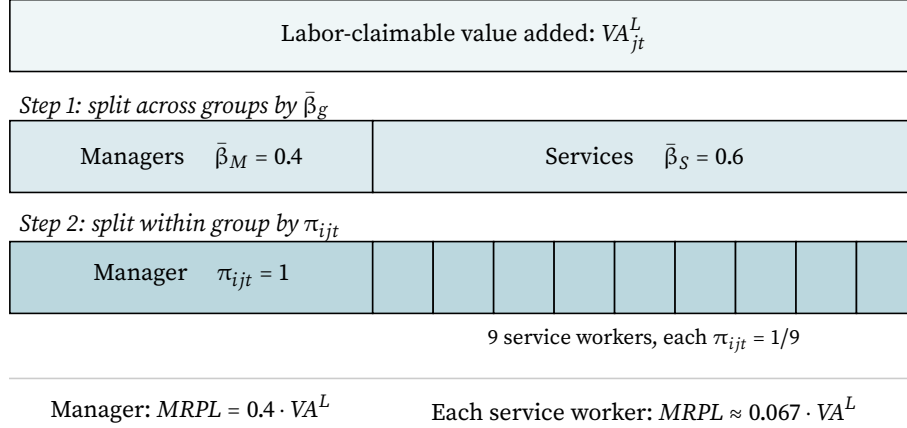
Equation (5) is therefore always a valid accounting allocation. Section 2.2 states when it also equals the annual claim associated with a worker's hourly marginal product.

The adding-up property also delivers the firm-year benchmark. Let annual worker earnings be W_{ijt} and let the firm-year wage bill be $W_{jt} \equiv \sum_{i \in j,t} W_{ijt}$. The worker-level construction then implies the aggregate labor-claim markdown

$$(7) \quad \mu_{jt}^{\text{agg}} \equiv \frac{\sum_{i \in j,t} MRPL_{ijt}^{\text{alloc}}}{\sum_{i \in j,t} W_{ijt}} = \frac{VA_{jt}^L}{W_{jt}}.$$

Equation (7) is the exact bridge from the worker-level construction to the firm-year labor-claim benchmark. Equivalence between this benchmark and a production-function markdown requires the same restrictions that make VA_{jt}^L the total marginal-product claim of labor.

FIGURE 1. Two-step allocation of labor-claimable value added: an illustrative calibration



Note: Illustration normalizes VA_{jt}^L to one. The first step allocates the labor residual across groups using contribution weights $\bar{\beta}_g$; the second step allocates each group claim across workers using within-group effective-hours shares π_{ijt} . Calibration: one manager and nine service workers with identical effective hours within the service group; $\bar{\beta}_M = 0.4$, $\bar{\beta}_S = 0.6$.

Human-capital weights and the markdown. The empirical baseline sets q_{it} with a transparent Mincer-style calibration. This choice borrows the standard earnings-function variables, schooling and potential experience, from the human-capital literature (Mincer 1974; Heckman, Lochner, and Todd 2006), but uses them only as predetermined productivity shifters in the allocation of value added. The calibration is therefore a convention, not a direct estimate of each worker's productivity.

$$(8) \quad q_{it} = \exp \left[\rho_s (s_i - 6) + 0.03 x_{it}^{\text{pot}} - 0.0005 (x_{it}^{\text{pot}})^2 \right],$$

where s_i is assigned years of schooling and $x_{it}^{pot} = \min\{\max(\text{age}_{it} - s_i - 6, 0), 40\}$ is capped potential experience. The baseline uses $\rho_s = 0.07$ and maps the three education groups to $s_i \in \{6, 12, 17\}$. The schooling slope is deliberately moderate: it is below the global private-return benchmark summarized by Psacharopoulos and Patrinos (2018) and in the range of Portuguese returns documented by Campos and Reis (2018). The experience profile is concave and restrained; the cap prevents late-career age differences from dominating the within-cell allocation.

This design avoids two unattractive extremes. Raw hours, $q_{it} = 1$, avoid wage circularity but impose equal product per hour within broad occupation cells. Wage-residual productivity weights allow worker heterogeneity but use the same wage measure that later appears in the markdown denominator. Under the wage-setting benchmark, wages combine productivity and markdowns, so wage-based productivity weights can mechanically move wage-setting differences into the numerator. The human-capital-hours baseline keeps wages out of the allocation while allowing education and potential experience to shift efficiency per hour. The raw-hours specification remains a transparent robustness check.

Let W_{ijt} denote annual earnings. The allocated worker-level markdown divides the allocated annual labor claim by annual earnings:

$$(9) \quad \mu_{ijt}^{\text{alloc}} \equiv \frac{MRPL_{ijt}^{\text{alloc}}}{W_{ijt}}.$$

The same object can be written as the hourly allocated marginal product divided by the hourly wage W_{ijt}/h_{ijt} when hours and earnings are measured over the same period.

A worked example fixes ideas. Continue the calibration in Figure 1, with VA_{jt}^L normalized to one, so the manager's allocated claim is 0.40 and each of the nine service workers receives about 0.067. Suppose the manager earns 0.25 and each service worker earns 0.05 in the same units. The manager's markdown is then $0.40/0.25 = 1.6$ and each service worker's is $0.067/0.05 \approx 1.33$: both are paid less than the labor claim they generate, and the manager bears the larger wedge. The wage-weighted aggregate is VA_{jt}^L divided by the 0.70 wage bill, about 1.43, consistent with the firm-year benchmark in (7). In logs,

$$\ln \mu_{ijt}^{\text{alloc}} = \ln VA_{jt}^L + \ln \bar{\beta}_{g(i),s(j)} + \ln q_{it} + \ln h_{ijt} - \ln \left(\sum_{k \in g(i),j,t} q_{kt} h_{kjt} \right) - \ln W_{ijt}.$$

This is the baseline measure: an accounting-consistent markdown built from value added net of capital costs, group contribution weights, human-capital-adjusted hours, and annual earnings. It is not, by itself, a structural parameter. The next section gives conditions under which it can be read as a marginal-product markdown; Section 2.5 explains how the fixed-effect components should be interpreted.

2.2. Production benchmark

This subsection states the production restrictions that turn the accounting allocation into a marginal-product measure. It first gives the exact Cobb-Douglas benchmark and then a weaker separable extension that preserves within-firm-year relative marginal products. In what follows, the superscript “alloc” marks the constructed allocation, “true” the underlying structural object, and a leading h an hourly rather than annual magnitude.

ASSUMPTION 1 (Labor-claimable residual as labor block). *The residual VA_{jt}^L in (1) equals the labor block of firm-year production revenue after capital is paid its user cost. Product-market rents, fixed costs, unmeasured capital services, taxes or subsidies, and capital measurement error either are absent or enter as firm-year factors that do not change the within-firm allocation rule across workers.*

ASSUMPTION 2 (Cobb-Douglas aggregation across labor groups). *Labor-claimable value added can be written as*

$$VA_{jt}^L = A_{jt} \prod_{g \in \mathcal{G}_{jt}} L_{gjt}^{\bar{\beta}_{g,s(j)}}, \quad \sum_{g \in \mathcal{G}_{jt}} \bar{\beta}_{g,s(j)} = 1.$$

ASSUMPTION 3 (Additive effective-hours aggregation within group). *Within each labor group,*

$$L_{gjt} = \sum_{i \in g, j, t} q_{it} h_{ijt},$$

where $h_{ijt} \geq 0$ is observed annual hours and $q_{it} > 0$ is the calibrated efficiency-per-hour weight in (8). For derivative and ratio statements, restrict attention to firm-year-group cells with $L_{gjt} > 0$ and worker observations with positive hours. Thus the derivative below is an hourly marginal product evaluated at observed hours, and the annual claim multiplies that hourly derivative by h_{ijt} .

Assumption 1 defines when the residual assigned to labor is the relevant production block. Assumptions 2 and 3 then turn the two-step allocation from accounting into production theory. These restrictions are stronger than the pure accounting identity in (6), but they make the interpretation of the allocated measure transparent.

Exact Cobb-Douglas interpretation. Under Cobb-Douglas aggregation across groups, each group receives a fixed share of the labor block, equal to its normalized output elasticity. Under additive aggregation within group, each worker’s annual claim is proportional to effective hours. The two-step allocation in (5) therefore equals the worker’s observed hours times the local derivative of firm value added with respect to those hours.

PROPOSITION 1 (The allocation equals the marginal-product claim under Cobb-Douglas). *Suppose Assumption 1 and Assumptions 2 and 3 hold. Consider an observed firm-year-group cell with positive effective labor, $L_{g(i)jt} > 0$. Then the two-step allocation coincides with the worker's marginal revenue product of labor: evaluated at observed hours, worker i 's hourly marginal product is*

$$\frac{\partial VA_{jt}^L}{\partial h_{ijt}} = MRPL_{ijt}^{h,true} = VA_{jt}^L \cdot \bar{\beta}_{g(i),s(j)} \cdot \frac{q_{it}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}}.$$

The annual allocated labor claim equals observed hours times that hourly marginal product:

$$MRPL_{ijt}^{alloc} = h_{ijt} MRPL_{ijt}^{h,true} = VA_{jt}^L \cdot \bar{\beta}_{g(i),s(j)} \cdot \frac{q_{it} h_{ijt}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}}.$$

In particular, (5) is both an exact accounting allocation and the exact annual claim associated with the worker's hourly marginal product.

The proof is in Appendix A.1. Proposition 1 is the benchmark interpretation for the framework. It is the proposition that turns the worker-level allocation from bookkeeping into production theory.

Separable proportionality. The benchmark proposition is useful, but it is stronger than what is needed for many within-firm comparisons. Suppose the residual assigned to labor is the relevant labor production block, and the nested labor aggregate above enters a separable, increasing outer transformation $\mathcal{H}(\cdot)$ (formalized in Assumption 4). Then the allocated measure need not equal the true marginal product in levels. But it still differs from the true marginal product only by a common firm-year scalar. Relative comparisons within firm-year remain intact.

ASSUMPTION 4 (Separable outer transformation). *Suppose labor-claimable value added can be written as*

$$VA_{jt}^L = \tilde{A}_{jt} \mathcal{H} \left(\prod_{g \in \mathcal{G}_{jt}} L_{gjt}^{\bar{\beta}_{g,s(j)}} \right),$$

where $\mathcal{H}(\cdot)$ is increasing and differentiable, and $\mathcal{H}(X_{jt}) > 0$ at observed inputs.

PROPOSITION 2 (Separable proportionality). *Suppose Assumption 1 and Assumptions 3 and 4 hold. Consider observations with positive hours in firm-year-group cells with $L_{g(i)jt} > 0$. Then the true hourly marginal product satisfies*

$$MRPL_{ijt}^{h,true} \equiv \frac{\partial VA_{jt}^L}{\partial h_{ijt}} = \xi_{jt} MRPL_{ijt}^{h,alloc},$$

where $MRPL_{ijt}^{h,alloc} \equiv MRPL_{ijt}^{alloc}/h_{ijt}$ and ξ_{jt} is a common firm-year scalar. Equivalently, the annual true claim equals $\xi_{jt}MRPL_{ijt}^{alloc}$. Hence, for any two workers i and k in the same firm-year with positive allocated hourly claims,

$$\frac{MRPL_{ijt}^{h,true}}{MRPL_{kjt}^{h,true}} = \frac{MRPL_{ijt}^{h,alloc}}{MRPL_{kjt}^{h,alloc}}, \quad \frac{\xi_{jt}MRPL_{ijt}^{alloc}}{\xi_{jt}MRPL_{kjt}^{alloc}} = \frac{MRPL_{ijt}^{alloc}}{MRPL_{kjt}^{alloc}}.$$

The proof is in Appendix A.2. The scalar ξ_{jt} is the local elasticity of the outer transformation $\mathcal{H}$ at the firm-year input bundle, a returns-to-scale term that equals one when $\mathcal{H}$ is the identity and Proposition 1 applies. Proposition 2 is weaker than Proposition 1. It says that the allocated measure remains informative about within-firm relative marginal products even when the exact level interpretation fails, because the common scalar cancels from any within-firm-year ratio.

2.3. Wage-setting benchmark

The production block tells us what the worker contributes. To interpret the gap between that contribution and earnings as labor market power, the framework needs a wage-setting block. We use an own-earnings labor-supply benchmark. Its role is not to recover the full labor-supply system inside the firm, but to state the conditions under which the worker-level markdown has the familiar monopsony interpretation (Robinson 1933; Manning 2003).

ASSUMPTION 5 (Separable worker-firm labor supply). *Each worker faces an own-earnings expected-employment schedule $\ell_{ijt}(W_{ijt}; z_{ijt})$ that depends only on annual earnings W_{ijt} and on a vector z_{ijt} of non-wage shifters. In particular, ℓ_{ijt} does not depend on earnings offered to other workers at the same firm.*

Assumption 5 rules out cross-earnings effects across workers. This is a strong separability restriction. It rules out wage ladders, internal equity constraints, and substitution in retention across workers. The restriction is useful because it lets the local firm problem collapse to worker-by-worker first-order conditions. Without it, the markdown-elasticity relationship becomes a system involving the full matrix of cross-earnings labor-supply elasticities.

Fix a firm-year and take the vector of true annual worker-level claims $\{MRPL_{ijt}^{true}\}_{i \in j,t}$ as given, where $MRPL_{ijt}^{true} \equiv h_{ijt}MRPL_{ijt}^{h,true}$ is the annual counterpart of the hourly product in Propositions 1 and 2. This is the standard monopsony shortcut: the firm treats $MRPL^{true}$ as locally fixed when varying a single earnings offer. The resulting wedge therefore follows from a local, worker-by-worker first-order condition. It does not internalize how raising

worker i 's earnings changes coworkers' marginal products through the additive aggregator in Assumption 3. Under Assumption 5, the local earnings problem is

$$(10) \quad \max_{\{W_{ijt}\}_{i \in j, t}} \sum_{i \in j, t} (MRPL_{ijt}^{\text{true}} - W_{ijt}) \ell_{ijt}(W_{ijt}; z_{ijt}).$$

The intuition is simple. If the firm raises a worker's earnings, it hires or retains that worker more easily. At the optimum, the firm trades off a lower surplus per filled position against a higher employment probability. The first-order condition implies a markdown wedge that is a simple function of the labor-supply elasticity faced for that worker.

PROPOSITION 3 (Static worker-level wage-setting wedge). *Suppose (10) has an interior solution and each $\ell_{ijt}(W_{ijt}; z_{ijt})$ is differentiable. At the solution, suppose $W_{ijt} > 0$, $\ell_{ijt} > 0$, and $\partial \ell_{ijt} / \partial W_{ijt} > 0$. Define the true worker-level markdown as*

$$\mu_{ijt}^{\text{true}} \equiv \frac{MRPL_{ijt}^{\text{true}}}{W_{ijt}},$$

and define the worker-specific labor-supply elasticity to the firm as

$$\eta_{ijt} \equiv \frac{\partial \ell_{ijt}}{\partial W_{ijt}} \frac{W_{ijt}}{\ell_{ijt}}.$$

Then the firm's first-order condition implies

$$(11) \quad \mu_{ijt}^{\text{true}} = 1 + \frac{1}{\eta_{ijt}}.$$

The proof is in Appendix A.3. Proposition 3 is the bridge from production to monopsony: the markdown is governed entirely by the worker-specific labor-supply elasticity to the firm, the elasticity that the empirical monopsony literature estimates directly and that Sokolova and Sorensen (2021) summarize across settings. The wedge falls monotonically as that elasticity rises and collapses to the competitive benchmark, $\mu_{ijt}^{\text{true}} \rightarrow 1$, as $\eta_{ijt} \rightarrow \infty$, so a less replaceable worker, with a lower elasticity of supply to the firm, carries a larger markdown. Under Proposition 1, the allocated markdown equals the true markdown, so

$$\mu_{ijt}^{\text{alloc}} = \mu_{ijt}^{\text{true}} = 1 + \frac{1}{\eta_{ijt}}.$$

Under Proposition 2, the true markdown still satisfies (11), while the allocated markdown

differs only by the common firm-year scalar:

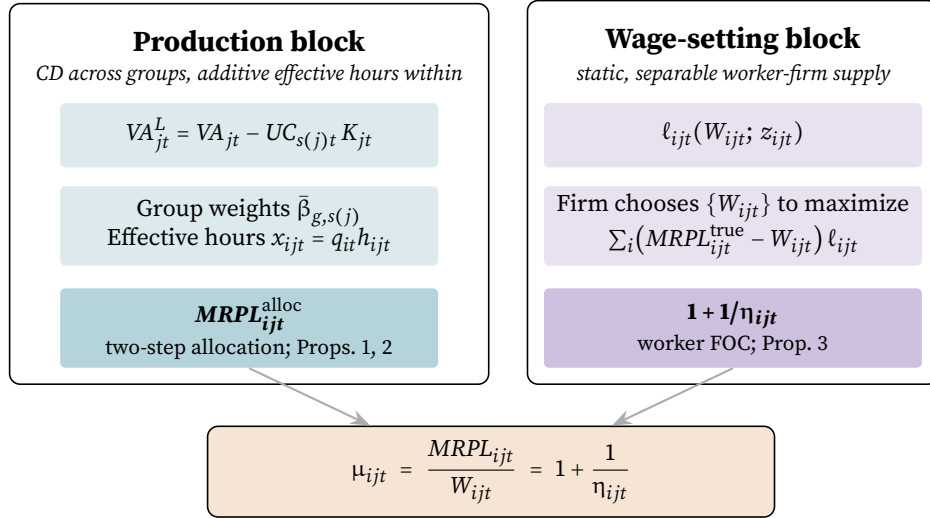
$$(12) \quad \mu_{ijt}^{\text{alloc}} = \frac{1}{\xi_{jt}} \left(1 + \frac{1}{\eta_{ijt}} \right).$$

The own-earnings benchmark is deliberately narrow. If expected employment depends on the full earnings vector, $\ell_{ijt} = \ell_{ijt}(\mathbf{W}_{jt}; z_{ijt})$, the first-order condition for earnings W_{mjt} is

$$\ell_{mjt} = \sum_{i \in j, t} \left(MRPL_{ijt}^{\text{true}} - W_{ijt} \right) \frac{\partial \ell_{ijt}}{\partial W_{mjt}}.$$

Equivalently, markdowns depend on the full matrix of own- and cross-earnings labor-supply elasticities, not only on η_{ijt} . Recovering that matrix would require a richer labor-supply or assignment model. The empirical exercise below does not identify those cross-earnings responses, so we use Proposition 3 only as an own-earnings benchmark for interpreting the allocated markdown.

FIGURE 2. Production block, wage-setting block, and the worker-level markdown



$$\sum_i MRPL_{ijt}^{\text{alloc}} = VA_{jt}^L \implies \mu_{jt}^{\text{agg}} = VA_{jt}^L / W_{jt}$$

Note: The production block delivers the allocated worker-level marginal-product claim; the wage-setting block delivers the markdown wedge. Allocated claims aggregate exactly to VA_{jt}^L , so the wage-weighted aggregate markdown equals VA_{jt}^L / W_{jt} . The displayed equality $\mu_{ijt} = 1 + 1/\eta_{ijt}$ holds exactly under Proposition 1; under Proposition 2 it holds up to the firm-year scalar ξ_{jt} in (12).

2.4. Markdown decomposition as a descriptive map

The structural benchmark gives the measure a clear interpretation. Under Propositions 1 and 3, $\ln \mu_{ijt}^{\text{alloc}} = \ln(1 + 1/\eta_{ijt})$. Under Propositions 2 and 3, it differs from that wage-setting wedge only by the additive firm-year term $-\ln \xi_{jt}$. The fixed-effect system below therefore organizes variation in constructed markdowns, but it still does not identify the underlying elasticities or their equilibrium determinants.

For compactness, write μ_{ijt} for the empirical markdown used in the decomposition. In the baseline, this is the human-capital-adjusted-hours allocated markdown in (9).

Baseline firm-worker decomposition. The main additive specification starts from the markdown analogue of a standard AKM system:

$$(13) \quad \ln \mu_{ijt} = \alpha_j + \chi_i + \tau_t + \varepsilon_{ijt},$$

where α_j is a firm component, χ_i is a worker component, τ_t is a year effect, and ε_{ijt} is a residual. These components summarize persistent sources of markdown variation in the constructed human-capital-adjusted-hours measure.

Occupation and market extensions. The occupation extension adds group effects:

$$(14) \quad \ln \mu_{ijt} = \alpha_j + \delta_{g(i)} + \chi_i + \tau_t + \rho \text{Tenure}_{it} + \varepsilon_{ijt},$$

The market extension adds industry-by-region-by-year effects:

$$(15) \quad \ln \mu_{ijt} = \kappa_{rt} + \alpha_j + \delta_{g(i)} + \chi_i + \rho \text{Tenure}_{it} + \varepsilon_{ijt},$$

where κ_{rt} is a local-market-by-year component and r indexes an industry-by-region labor market.

These equations assign markdown variation to persistent firm, worker, group, and market components under an additivity restriction.

Variance accounting. For any fitted additive components, we report the variance identity

$$(16) \quad \text{Var}(y) = \sum_m \text{Var}(c_m) + 2 \sum_{m < n} \text{Cov}(c_m, c_n) + \text{Var}(\varepsilon),$$

where $y = \ln \mu_{ijt}$ and c_m are fitted components. Covariances are part of the variance decomposition. They describe whether firm, worker, occupation, and market markdown components reinforce or offset one another in the observed allocation.

Firm-year fixed effects under separability. Specification (13) is appropriate under Proposition 1, where the allocated and true markdowns coincide. Under Proposition 2, however, the allocated markdown carries a firm-year scalar:

$$\ln \mu_{ijt}^{\text{alloc}} = \ln \mu_{ijt}^{\text{true}} - \ln \xi_{jt}.$$

The additive structure $\alpha_j + \tau_t$ removes only the part of $\ln \xi_{jt}$ that can be written as a firm effect plus a year effect; any genuinely firm-year-specific component remains in α_j and in the residual. To remove $\ln \xi_{jt}$ exactly, the natural specification is

$$(17) \quad \ln \mu_{ijt} = \alpha_{jt} + \chi_i + \varepsilon_{ijt},$$

which collapses the firm and time effects into a single firm-year intercept.

Specification (17) has a clear benefit and a clear cost. The benefit is that the separability scalar ξ_{jt} drops out of χ_i and ε_{ijt} entirely, so the within-firm-year variation of the allocated markdown equals the within-firm-year variation of the true markdown:

$$\chi_i + \varepsilon_{ijt} = \ln \mu_{ijt}^{\text{true}} - \mathbb{E}[\ln \mu_{ijt}^{\text{true}} \mid j, t].$$

Movers identify the worker component χ_i under standard connected-set arguments, and the firm-year intercept separates χ_i from firm-year shocks, including ξ_{jt} . The cost is that α_{jt} combines the firm-year mean of $\ln \mu_{ijt}^{\text{true}}$ with $-\ln \xi_{jt}$. Cross-firm or cross-year comparisons of markdown levels therefore require either Proposition 1, which sets $\xi_{jt} \equiv 1$, or an external restriction on how ξ_{jt} varies across (j, t) .

A second caveat is interpretive. Specification (17) delivers χ_i as the AKM projection of worker-level variation in $\ln \mu_{ijt}^{\text{alloc}}$ onto an additive worker effect. For χ_i to be interpreted as the structural worker-specific part of the wage-setting wedge, one must also assume that $\ln(1 + 1/\eta_{ijt})$ itself decomposes additively in a worker effect and a firm-year effect. This is a restriction on the labor-supply elasticity η_{ijt} that is not implied by the production block. Without it, χ_i remains a useful descriptive summary of portable worker-level markdown heterogeneity rather than a structural elasticity parameter.

In this draft, (17) remains a diagnostic exercise rather than the main specification. The main additive evidence reports the long-run firm-worker-year decomposition in (13), the occupation and market extensions below, and auxiliary comparisons between firm-year and firm-year-occupation specifications. The firm-year-occupation version matters because the denominator in (4) is defined within firm-year-group cells; a firm-year intercept removes ξ_{jt} , but it need not remove denominator variation that differs across occupation groups within the same firm-year.

Why not a match-effect decomposition. A natural way to look for non-additivity is to add a persistent worker-firm match intercept. We do not make that regression part of the main evidence. Once a match intercept absorbs persistent worker-firm variation, the remaining variance accounting is hard to read: gross shares and covariance terms become large, and the tenure coefficient is identified from thin within-match variation. A match intercept also mixes several objects that the paper cannot separately identify: match surplus, firm pay policy, worker selection, amenities, and measurement error in the allocated claim. The BLM exercise in Section 5 is therefore the preferred non-additive diagnostic. It asks a narrower question: in a grouped worker-type and firm-class model, how much explanatory power is gained by allowing type-class interactions beyond additive worker-type and firm-class terms?

2.5. Interpreting the decomposition

The fixed effects decompose the allocated markdown, not a complete equilibrium model of search, bargaining, amenities, or welfare. The framework gives this measure a marginal-product and wage-setting interpretation under explicit benchmarks. Outside those benchmarks, the decomposition remains useful as an accounting-consistent description of how markdown dispersion is divided across firms, workers, occupation groups, local markets, and their covariances.

Firm and worker components. The firm effect α_j in (13) captures persistent between-firm variation in markdowns conditional on the maintained production-side allocation. Under the wage-setting benchmark, it can be read as persistent firm-level variation in wage-setting wedges. Under separability (Proposition 2), α_j also includes the part of the firm-year scalar ξ_{jt} that can be written as a firm effect, while genuinely firm-year-specific components remain in the residual. Replacing $\alpha_j + \tau_t$ with α_{jt} removes ξ_{jt} exactly, but then firm-year levels combine true markdown levels with the separability scalar.

The worker effect χ_i captures persistent worker-specific variation in the constructed markdown. Under Proposition 3, this variation can reflect worker-level differences in labor-supply elasticities to the firm. Under (17), χ_i is separated from firm-year shocks, including ξ_{jt} ; under (13), firm-year-specific shifts can be reflected in χ_i through non-random mobility. Treating χ_i as a structural elasticity parameter would require an additional additivity restriction on η_{ijt} and evidence that the effective-hours convention has not assigned markdown-relevant worker heterogeneity to productivity.

Occupation and market components. The group effect δ_g is interpreted conditional on the maintained group contribution weights $\bar{\beta}_g$ and the effective-hours weights q_{it} used in the allocation. A non-zero δ_g therefore says that allocated markdowns differ across groups

after imposing those measured productivity weights. It does not claim that all productivity differences have been removed without assumptions. The market effect κ_{rt} captures variation common to workers in the same industry-by-region market-year, and can reflect common differences in competition, demand, search frictions, or labor-supply elasticities. These components locate common markdown variation in the additive projection; by themselves, they do not identify the mechanism behind that variation.

Tenure and covariance terms. The baseline effective-hours weight in (8) uses potential experience, not firm-specific tenure. The coefficient ρ in (14) and (15) therefore describes residual tenure-related variation in constructed markdowns after the Mincer-style experience adjustment. Under Proposition 3, this residual can reflect tenure-related changes in labor-supply elasticities, firm-specific learning, or lock-in; distinguishing among those channels requires separate evidence.

Covariance terms are part of the economic description rather than a nuisance. They show whether firm, worker, occupation, and market markdown components reinforce or offset one another in the observed allocation. Their signs should not be imported mechanically from wage decompositions: markdowns compare wages with an allocated labor claim, so a positive or negative firm-worker covariance in markdowns is not, by itself, a standalone sorting result.

Scope of the interpretation. The baseline measure is the allocated markdown μ_{ijt}^{alloc} . Exact equality with a worker-level marginal product holds under Proposition 1; under separability, the measure is scaled by a firm-year term. The residual $VA_{jt}^L = VA_{jt} - UC_{s(j)t}K_{jt}$ is likewise a maintained measure of the value added assigned to labor. Product-market rents, fixed costs, unmeasured capital services, and capital measurement error can shift its level and therefore affect between-firm comparisons, while within-firm-year relative markdowns are less exposed because the common residual enters all workers proportionally. Recent evidence underscores this caution: Faia, Lochner, and Schoefer (2026) find that firm-level markdowns respond far less to the 2015 German minimum wage than a monopsony model predicts, which suggests that measured markdowns partly reflect distortions or measurement other than wage-setting power.

The empirical sections therefore use the fixed-effect decomposition as a descriptive map. Stronger mechanism claims come from the benchmark propositions or from later exercises that are explicitly presented as correlational diagnostics. The separability-scaled markdown in (A1) and any wage-based productivity convention belong in that sensitivity category: they are useful checks, but they should not be read as structural without the additional restrictions needed to identify the relevant firm-year scalar or to rule out wage markdowns inside the productivity measure.

3. Data and measurement

The empirical analysis uses matched Portuguese administrative records from *Quadros de Pessoal* (QP) and *Sistema de Contas Integradas das Empresas* (SCIE). The unit of observation is a worker–firm–year. QP supplies worker earnings, hours, occupation, education, age, tenure, firm identifiers, industry, and region. SCIE supplies firm balance sheets and income statements. The estimation sample covers 2004–2024, the years in which the worker records can be linked to cleaned SCIE firm-year accounts and to the production-function inputs needed to build worker-level markdowns.

This section documents how the measure defined in Section 2 is built in the data.

3.1. Data sources and sample construction

QP is a mandatory employer survey that records the workforce of Portuguese firms. The cleaning procedure harmonizes firm identifiers across years, keeps one primary job per worker-year, maps education into three groups, harmonizes qualification and occupation codes across classification changes, constructs annual earnings from wage components and months worked, and constructs annual hours. The worker panel keeps workers with internally consistent age histories and ages 16–78. SCIE provides the firm accounts used to measure value added, investment, capital, depreciation, interest expenses, sales, industry, and employment. Monetary variables are deflated to 2010 euros.

Appendix Table B1 reports the main sample screens. The cleaned QP worker-year panel contains 71.2 million observations for 9.1 million workers and 1.0 million firms over 1985–2024. Linking to cleaned SCIE firm-years leaves 40.3 million observations over 2004–2024. Requiring the inputs needed for the baseline allocated marginal product leaves 28.4 million observations. The additive decomposition then uses the largest connected worker-firm set after requiring positive finite markdowns, non-missing occupation group, and non-missing tenure. This connected set contains 13.7 million observations, 2.4 million workers, and 166,035 firms.

The row labeled “worker-year with individual quality” is reported for transparency because we also build a wage-based worker-quality diagnostic from entry wages. It is not a baseline restriction. The preferred markdown uses calibrated human-capital-adjusted hours rather than entry-wage residuals.

3.2. Labor-claimable value added

We construct the residual in (1) from SCIE firm accounts. The procedure builds real capital with a forward and backward perpetual-inventory method. It forms a raw user-cost rate by dividing depreciation plus interest expenses by the capital stock—taking interest in absolute value, since SCIE records it with a negative sign—then winsorizes that ratio

within sector-year cells and uses the sector-year median. When depreciation data are sparse before 2010, the procedure carries the sector’s 2010 depreciation and implied-return components back to earlier years. Negative depreciation entries are set to missing before this calculation.

The baseline labor-claimable value added measure is market value added less a cleaned user-cost flow for capital. The main alternative subtracts the raw depreciation-plus-interest capital-cost flow, with a pre-2010 median fallback when the depreciation information is sparse. All main exhibits use the cleaned user-cost measure unless the text explicitly labels a sensitivity.

3.3. Occupation weights and human-capital hours

Workers are assigned to three broad qualification groups: service/manual workers, technicians/clerks, and managers/professionals. We aggregate QP workers to firm-year-by-occupation inputs and classify each firm-year by its *input pattern*—the subset of the three groups that the firm-year employs. The group weights $\beta_{g,s(j)}$ in (2) come from sector-specific production functions estimated for labor-claimable value added.

We estimate these production functions with the proxy-variable estimator of Akerberg, Caves, and Frazer (2015), separately for each sector and input pattern and pooling across years. Labor-claimable value added is treated as a function of the group labor inputs alone, scaled by Hicks-neutral productivity: capital does not enter the production function because its user cost is already removed in (1), and intermediate inputs (materials plus purchased services) serve only as the proxy. The first stage projects $\ln VA_{jt}^L$ on a cubic polynomial in the group labor inputs and the intermediate-input proxy. The second stage recovers the labor elasticities $\beta_{g,s(j)}$ from an AR(1) productivity process, identifying them under the assumption that a firm’s current employment is predetermined relative to its productivity innovation, so that the innovation is orthogonal to current labor inputs. The baseline uses group headcounts as the labor inputs; hours, human-capital-adjusted hours, and the group wage bill are alternatives reported as sensitivities. For each sector–pattern cell, the group weight is the group’s labor elasticity divided by the sum of labor elasticities across the groups the cell employs, matching the normalized weights in (2).

Within each firm-year-group cell, the baseline uses the effective-hours index in (8). Annual hours are first winsorized within year at the 2.5th and 97.5th percentiles. Schooling years are mapped from the three education groups to 6, 12, and 17 years; potential experience is capped at 40 years; and firm tenure is excluded from the baseline weight. A worker’s within-cell allocation share is then effective hours divided by total effective hours in the same firm-year-group cell.

This choice keeps the worker’s own wage out of the numerator-side allocation.

3.4. Worker-level markdowns

The baseline marginal-product claim follows (5), with production weights indexed by sector and occupation-input pattern. The denominator in (9) is annual earnings from QP. Before the decomposition, the baseline markdown is winsorized at the 2.5th and 97.5th percentiles, restricted to positive finite values, logged, and then restricted to the connected worker-firm set.

We construct six allocated markdown variants. The headline measure uses the cleaned labor-claimable value-added numerator, headcount-based production-function labor elasticities, and Mincer-adjusted effective hours. A raw-hours variant keeps the same numerator and production weights but allocates within firm-year-group cells using raw hours. An effective-hours production-function variant also estimates production elasticities with effective-hours group inputs. Three additional variants repeat these choices using the alternative capital-cost subtraction. The main text reports the headline measure and uses the other variants only as labeled sensitivities.

3.5. Descriptive statistics

Table 1 describes the connected-set sample used in the main decompositions. Technicians and clerks account for 70 percent of observations. Service/manual workers account for 19 percent, and managers/professionals account for 11 percent. Mean annual earnings rise sharply across groups: 7,776 euros for service/manual workers, 13,097 euros for technicians/clerks, and 26,886 euros for managers/professionals. Tenure is lower for service/manual workers than for the other two groups.

TABLE 1. Descriptive statistics by occupation group

Occupation group	N	Share	Annual wage		Tenure
			Mean	SD	
Service/Manual Workers	2,703,103	0.19	7,776	5,077	4.3
Technicians & Clerks	9,571,753	0.70	13,097	8,938	7.0
Managers & Professionals	1,450,047	0.11	26,886	17,300	6.6

Note: N counts worker-year observations in the connected set used in the main additive decompositions. Annual wages are deflated to 2010 euros; tenure is measured in years. Occupation groups follow the three-category aggregation used throughout the paper.

3.6. Markdown distribution

Table 2 reports the baseline markdown distribution. The median service/manual worker has a markdown of 0.95, which corresponds to a wage slightly above the allocated claim. The median technician/clerk has a markdown of 1.75. The median manager/professional

has a markdown of 1.93. Appendix Table B2 shows that the raw-hours allocation barely changes group means and medians: the largest median change is 0.02 log points. The occupation ranking is therefore not driven by the Mincer adjustment inside firm-year-group cells.

TABLE 2. Distribution of estimated individual markdowns by occupation group

Baseline distribution of estimated markdowns (log units unless noted)						
Occupation group	N	Mean	Median	IQR	Median μ	Median $w/MRPL$
Service/manual workers	2,703,103	-0.02	-0.05	[-0.90, 0.75]	0.95	1.05
Technicians/clerks	9,571,753	0.57	0.56	[0.15, 1.02]	1.75	0.57
Managers/professionals	1,450,047	0.70	0.66	[0.13, 1.30]	1.93	0.52

Note: The table reports moments of $\ln \mu_{ijt}^{\text{alloc}}$ by occupation group. The median μ column exponentiates the median log markdown; the wage-share column reports its inverse, $W_{ijt}/MRPL_{ijt}^{\text{alloc}}$.

Between and within firm-years. Most occupation markdown gaps come from sorting across firm-years rather than from differences within the same firm-year. Table 3 decomposes each mean gap relative to service/manual workers. For technicians/clerks, the total gap is 0.59 log points, of which 0.46 comes from employment in higher-markdown firm-years and 0.12 remains within firm-year. For managers/professionals, the total gap is 0.72 log points, with 0.50 between firm-years and 0.22 within firm-year. The allocation of workers across firms is therefore central to the occupation gradient in markdowns, but within-firm-year differences remain economically meaningful, especially for managers/professionals.

TABLE 3. Between and within firm-year decomposition of occupation markdown gaps

Occupation group	Total gap	Between firm-year	Within firm-year	Between share (%)
Technicians/clerks	0.59	0.46	0.12	79.0
Managers/professionals	0.72	0.50	0.22	70.0

Note: The table decomposes mean log-markdown gaps relative to service/manual workers. The between-firm-year component is the difference across occupation groups in average firm-year log markdowns. The within-firm-year component is the difference in deviations from those firm-year means. Components sum to the total gap up to rounding.

4. Additive decomposition

The first empirical exercise decomposes log-markdown dispersion using additive fixed-effect projections. Three facts organize the section. Firm effects are the largest KSS-corrected component in the baseline AKM system, so markdowns differ sharply across

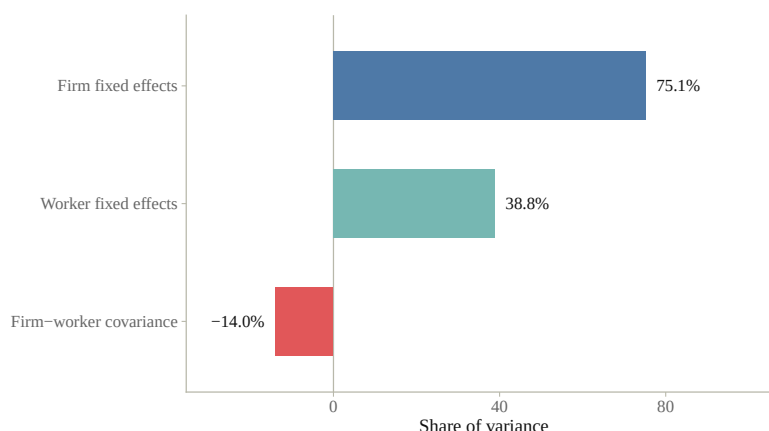
employers even after holding fixed worker composition. Worker effects are also large, so portable worker-level differences matter for wage-setting wedges. The firm-worker covariance is negative, which means the two components offset each other in the observed allocation rather than simply stacking on the same workers.

The section uses two complementary variance decompositions. For the baseline firm-worker specification, the headline shares are KSS-corrected for limited-mobility bias using the leave-out estimator of Kline, Saggio, and Sølvssten (2020). For the occupation and local-market extensions, the figures report the full fixed-effect variance identity because it shows how much occupation effects, market effects, residuals, and covariances contribute. The corresponding KSS corrections are reported in the text as firm-worker diagnostics after partialling out the additional controls.

4.1. AKM baseline

The AKM baseline in (13) decomposes markdowns into persistent firm and worker components on the connected set. Figure 3 shows the KSS-corrected variance shares. Firm effects contribute 75.1 percent of connected-set variance, worker effects contribute 38.8 percent, and twice the firm-worker covariance contributes -14.0 percent. The firm and worker shares need not sum to one hundred percent on their own, because the covariance term is also part of the variance identity.

FIGURE 3. KSS-corrected AKM baseline variance decomposition



Note: Bars are KSS-corrected shares of connected-set variance in $\ln \mu_{ijt}^{\text{alloc}}$. The AKM specification includes firm and worker effects with year controls; the covariance bar reports $2 \text{Cov}(\hat{\alpha}_j, \hat{\chi}_i)$.

The economic message is not that either firms or workers alone explain markdowns. The firm component says that some employers pay systematically lower wages relative to allocated claims than other employers do. The worker component says that some workers carry persistently higher markdowns across employers, consistent with heterogeneity

in outside options, mobility, bargaining, or remaining measurement error. The negative covariance says these two sources of dispersion partly offset: high worker-component markdowns are not concentrated in the highest firm-component markdown jobs.

Appendix Table C4 reports the uncorrected comparison and leverage diagnostics. The correction leaves the firm share close to the uncorrected estimate but lowers the worker share and makes the covariance less negative. The large firm component is therefore not an artifact of limited mobility, while the worker component and the covariance are more sensitive to leave-out correction.

Firm wage premia and firm markdown components. The markdown decomposition also reveals a new fact about high-wage firms. Standard wage decompositions show which firms pay more. They do not say whether those firms also generate larger labor claims, or how much of those claims workers receive. To connect wage premia to the production side, we estimate three fixed-effect regressions on the same connected sample:

$$(18) \quad y_{ijt} = \psi_j^y + \theta_i^y + \tau_t^y + u_{ijt}^y,$$

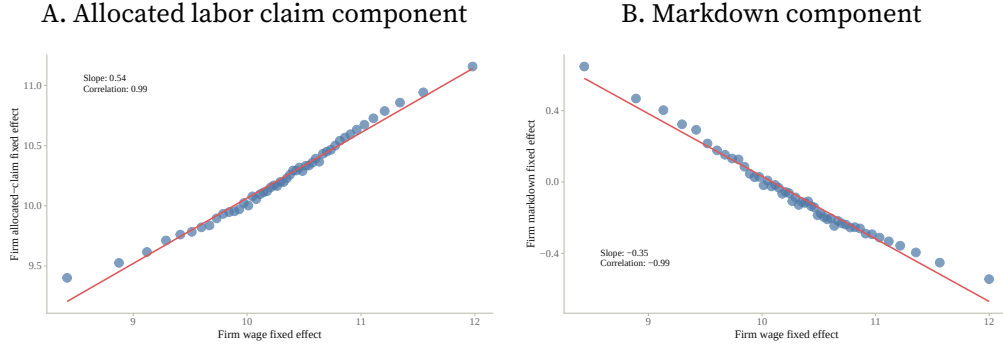
where y_{ijt} is log annual earnings, the log allocated labor claim, or the log markdown. The firm effect ψ_j^y is therefore a firm wage component, a firm claim component, or a firm markdown component, depending on the outcome. We estimate all three regressions on the same observations and with the same firm, worker, and year effects. This common specification makes the accounting transparent. The worker-level markdown equals the allocated claim divided by annual earnings. After centering the estimated firm effects, the firm markdown component is therefore the firm claim component minus the firm wage component.

Hours matter for this comparison. The allocated claim assigns the firm-year labor claim to workers using human-capital-adjusted hours: workers with more adjusted hours receive a larger share of the claim within their firm-year-occupation cell. Annual earnings are the wage denominator. The figure therefore does not compare a wage rate with a wage-rate measure of productivity, but rather annual earnings with an hours-based allocation of the firm's labor claim.

Figure 4 shows that high-wage firms are also high-claim firms. The correlation between the firm wage component and the firm allocated-claim component is 0.44. A one log point higher firm wage component is associated with a 0.57 log point higher firm claim component. These firms therefore sit higher in the distribution of labor claims, not only in the distribution of pay.

Panel B translates this fact into markdowns. High-wage firms have lower firm markdown components: the correlation is -0.35 , and the slope is -0.43 . This second slope is the accounting implication of Panel A. The markdown component equals the claim compo-

FIGURE 4. Firm wage premia, allocated labor claims, and markdown components



Note: Binscatters compare firm effects from (18). The horizontal axis is the firm component from the wage regression. Panel A plots the firm component from the allocated-claim regression. Panel B plots the firm component from the markdown regression. All firm effects are centered before plotting. The regressions use the connected-set worker-year sample; after singleton fixed-effect levels are removed, the plotted firm effects cover 154,369 firms.

ment minus the wage component, and the claim component rises by only 0.57 log points when the wage component rises by one log point. The economic content is therefore in Panel A: high-wage firms have higher allocated labor claims, but their wages are higher still. Workers at high-wage firms receive a larger share of the labor claim measured from the production side.

4.2. Occupation effects

The occupation-augmented specification in (14) adds fixed effects for the three occupation groups. This is a stringent comparison because the numerator of the markdown already allocates value added using occupation-specific production weights. A positive occupation effect therefore means that an occupation has a larger residual markdown after imposing those measured productivity weights, not merely that the occupation receives a larger share of value added.

Table 4 reports the regression. The tenure coefficient is -0.013 , so an additional year of tenure is associated with about a 1.3 percent lower markdown, conditional on the fixed effects. This gradient is statistically precise but small relative to fixed-effect dispersion; it should be read as residual tenure variation in the constructed markdown, not as a direct estimate of lock-in or learning.

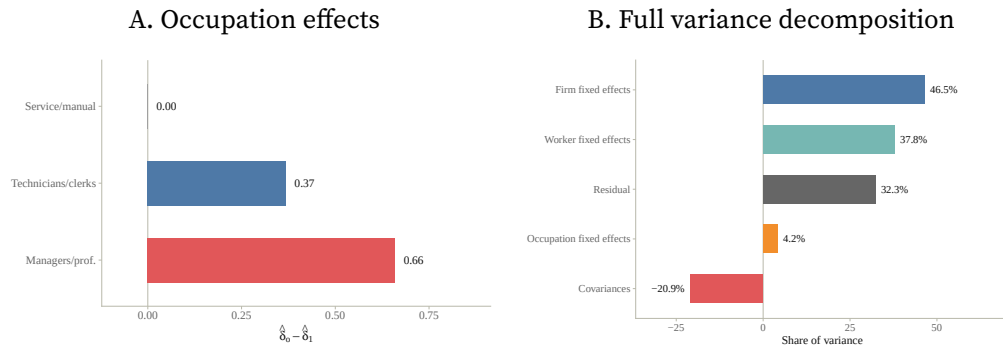
Figure 5 shows the occupation wedges and the full occupation-augmented variance decomposition. Firm effects explain 46.5 percent of total variance, worker effects 37.8 percent, occupation effects 4.2 percent, residual variation 32.3 percent, and all pairwise covariance terms together -20.9 percent. The KSS correction to the firm-worker part of this specification leaves the same ranking: after partialling out occupation, year, and tenure

TABLE 4. Occupation-augmented markdown decomposition

	(1)
A. Main coefficients	
<i>Dependent variable: Log markdown ($\ln \mu_{ijt}$)</i>	
Tenure	-0.013*** (0.0001)
B. Fixed effects and fit statistics	
Firm fixed effects	Yes
Worker fixed effects	Yes
Occupation fixed effects	Yes
Year fixed effects	Yes
Observations	13,283,092
R ²	0.68
Within R ²	0.002

Note: Dependent variable: $\ln \mu_{ijt}^{\text{alloc}}$. The specification includes firm, worker, occupation, and year fixed effects, plus tenure. Standard errors clustered at the worker level are in parentheses. Stars denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

FIGURE 5. Occupation-augmented components



Note: Panel A plots occupation fixed effects as $\hat{\delta}_0 - \hat{\delta}_1$, normalizing service/manual workers to zero. Panel B reports the full, uncorrected fixed-effect variance identity from (16); shares are percentages of total variance in $\ln \mu_{ijt}^{\text{alloc}}$. KSS-corrected firm-worker diagnostics for this specification are reported in Appendix Table C5.

controls, the corrected firm component contributes 75.0 percent, the corrected worker component contributes 41.9 percent, and twice the corrected firm-worker covariance contributes -16.9 percent.

The occupation wedge is 0.37 log points for technicians and clerks and 0.66 log points for managers and professionals, both relative to service and manual workers. In levels, those effects correspond to markdowns about 45 percent and 93 percent higher than the service/manual benchmark.

The occupation pattern is central for the paper. Managers and professionals earn much more in levels, but their allocated markdowns are also higher after imposing production

weights. In the wage-setting interpretation, wages rise with hierarchy, but not one-for-one with the allocated labor claim. This ordering matches the structural evidence in Jáñez and Delgado-Prieto (2025) that monopsony power, and its welfare cost, is larger for managers than for production workers in Portugal; we recover the same occupation ranking directly from measured markdowns.

4.3. Local labor markets

The local-market specification in (15) adds industry-by-region-by-year effects, while keeping occupation fixed effects separately. These components capture shocks and persistent differences common to workers in the same sector, region, and year: local demand, market thickness, institutions, and outside-option conditions can all be reflected in this term.

TABLE 5. Markdown decomposition with local labor market effects

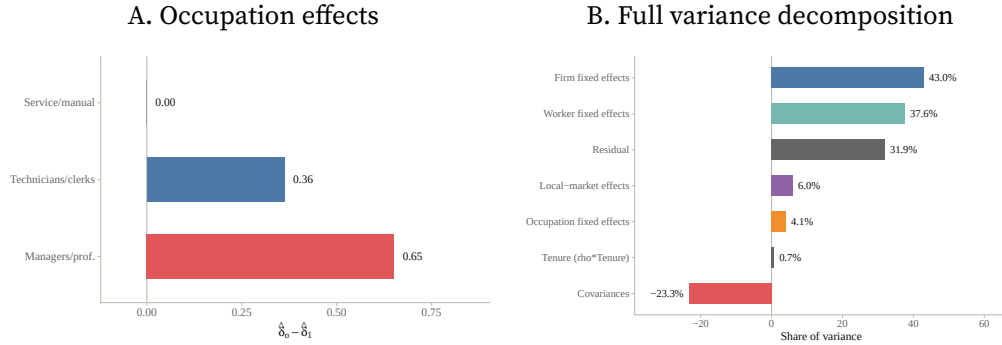
	(1)
A. Main coefficients	
<i>Dependent variable: Log markdown ($\ln \mu_{ijt}$)</i>	
Tenure	-0.012*** (0.0001)
B. Fixed effects and fit statistics	
Firm fixed effects	Yes
Worker fixed effects	Yes
LLM $\times$ year fixed effects	Yes
Occupation fixed effects	Yes
Observations	13,283,092
R ²	0.68
Within R ²	0.002

Note: Dependent variable: $\ln \mu_{ijt}^{\text{alloc}}$. Local labor market fixed effects are industry $\times$ region $\times$ year cells; occupation fixed effects enter separately. Standard errors clustered at the worker level are in parentheses. Stars denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

Table 5 reports the regression; the tenure coefficient remains negative, at -0.012 . Figure 6 shows that adding market components leaves the core ranking intact. Industry-by-region-by-year effects explain 6.0 percent of total variance. Firm effects account for 43.0 percent, worker effects for 37.6 percent, occupation effects for 4.1 percent, residual variation for 31.9 percent, and all pairwise covariances together for -23.3 percent. The KSS-corrected firm-worker diagnostic remains firm-heavy after partialling out tenure, occupation, and industry-by-region-by-year controls: firm effects contribute 78.8 percent, worker effects contribute 35.9 percent, and twice the firm-worker covariance contributes -14.7 percent.

The local-market exercise argues against a simple composition story. Market-year effects matter, but they do not eliminate the firm, worker, or occupation components.

FIGURE 6. Local-market components



Note: Panel A plots occupation fixed effects from the local-market specification, again normalized to the service/manual group. Panel B reports the full, uncorrected fixed-effect variance identity after adding industry $\times$ region $\times$ year effects; shares are percentages of total variance in $\ln \mu_{ijt}^{\text{alloc}}$. KSS-corrected firm-worker diagnostics are reported in Appendix Table C5.

The manager/professional occupation markdown remains 0.65 log points, and the technician/clerk markdown remains 0.36 log points. Thus the occupation ranking is not just a reflection of where different workers live or which industries employ them.

4.4. Covariances

Covariances are large and economically informative. Table 6 shows the uncorrected pairwise covariance terms in the occupation-augmented decomposition. The firm-worker covariance contributes -23.7 percent of total variance. Occupation-worker covariance contributes -2.7 percent, while firm-occupation covariance contributes 5.4 percent.

TABLE 6. Pairwise covariance terms in the occupation-augmented decomposition

Term	Covariance	Share of total variance (%)
Firm-Worker	-0.095	-23.7
Firm-Occupation	0.022	5.4
Occupation-Worker	-0.011	-2.7
Total covariances	-0.167	-20.9

Note: Each covariance term is $2 \text{Cov}(\cdot, \cdot)$ from the occupation-augmented variance identity in (16). Shares are percentages of total variance in $\ln \mu_{ijt}^{\text{alloc}}$ and are not KSS-corrected.

The negative firm-worker covariance means that high-worker-component markdowns and high-firm-component markdowns do not simply pile on top of one another. This is not the same as wage sorting. In the markdown measure, wages enter the denominator and allocated claims enter the numerator, so the covariance describes how persistent wage-setting wedges are arranged after the productivity allocation.

The positive firm-occupation covariance points in the opposite direction: firms with larger markdown components are more often paired with occupations that also have larger markdown components. The negative occupation-worker covariance partly offsets that reinforcement. Together, the covariance terms show why additive markdown decompositions need the full variance identity rather than a ranking of standalone fixed-effect variances. Before turning to non-additive complementarity, we ask whether the estimated market and firm components line up with standard local labor-market-power measures.

4.5. Connecting markdowns to market structure

The additive decompositions above organize markdown dispersion into components. We now relate selected components to market concentration and within-market firm size. These regressions are descriptive. They ask whether the estimated market and firm components move with standard measures of local labor market power.

Market concentration. For each occupation-specific local labor market, we estimate

$$(19) \quad \widehat{\kappa}_{rt}^o = \sum_o b_o \ln HHI_{rt}^o \cdot 1\{O = o\} + \gamma_t + \gamma_o + u_{rt}^o,$$

where HHI_{rt}^o is the employment concentration index in market-year (r, t) for occupation group o . Table 7 reports the results. The concentration slope is -0.144 for service/manual workers, 0.020 for technicians/clerks, and -0.048 for managers/professionals. The negative association is strongest for service/manual workers, the group for which local concentration is most likely to capture outside-option pressure. Measuring concentration within occupation groups is in the spirit of the task-based approach of Dodini et al. (2024), who show that accounting for the skills workers can carry across jobs reshapes measured labor-market concentration and the employer power it implies.

Within-market firm size. We next relate firm components to within-market payroll shares:

$$(20) \quad \widehat{\alpha}_j^o = b \ln Share_j + \gamma_{r(j)} + e_j,$$

where $Share_j$ is the firm's average payroll share in its local market and $\gamma_{r(j)}$ are market fixed effects. Table 8 shows a slope of -0.107 . Larger firms within a market have lower estimated firm markdown components. This pattern is not the naive prediction that larger employers always impose larger markdowns, as in models of size-based labor-market power (Jarosch, Nimczik, and Sorkin 2024); it is consistent instead with large firms paying more to attract and retain workers within the same market.

These market-structure regressions help interpret the additive components, but they

TABLE 7. Markdowns and market concentration

	(1)
A. Main coefficients	
<i>Dependent variable: Estimated market component (κ_{rt}^o)</i>	
Log concentration ($\ln HHI_{rt}^o$) $\times$ Service and Manual Workers	-0.144*** (0.015)
Log concentration ($\ln HHI_{rt}^o$) $\times$ Technicians and Clerks	0.020 (0.016)
Log concentration ($\ln HHI_{rt}^o$) $\times$ Managers and Professionals	-0.048** (0.019)
B. Fixed effects and fit statistics	
Year fixed effects	Yes
Occupation fixed effects	Yes
Observations	2,223
R ²	0.37
Within R ²	0.05

Note: Dependent variable: estimated occupation-specific local-market component κ_{rt}^o . Unit of observation is occupation group $\times$ industry $\times$ region $\times$ year; the regressor is log employment HHI in the same cell. Heteroskedasticity-robust standard errors are in parentheses. Stars denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

TABLE 8. Within-market firm size and markdowns

	(1)
A. Main coefficients	
<i>Dependent variable: Estimated firm component (α_j^o)</i>	
log(avg_shares)	-0.107*** (0.001)
B. Fixed effects and fit statistics	
Market fixed effects (industry $\times$ region)	Yes
Observations	277,224
R ²	0.04
Within R ²	0.02

Note: Dependent variable: estimated firm-by-occupation markdown component α_j^o . Unit of observation is a firm $\times$ occupation-group $\times$ industry-region cell. The regressor is the log of the firm's average payroll share in that local market; fixed effects are industry $\times$ region cells. Heteroskedasticity-robust standard errors are in parentheses. Stars denote $p < 0.10$, $p < 0.05$, and $p < 0.01$.

do not test whether the worker-firm structure itself is additive. They relate estimated component levels to familiar market measures. The next section turns to additivity directly.

5. Beyond additivity

The additive fixed-effect system imposes that a worker component has the same meaning at every firm and that a firm component has the same meaning for every worker. That

restriction is strong for markdowns. Bils, Kaymak, and Wu (2025) emphasize an economic setting in which monopsony power is tied to comparative advantage at the worker-firm level: firms have more scope to mark down wages when a worker is especially productive at that firm. In that case the relevant variation is not only a worker effect, a firm effect, or their covariance. It is also the interaction between worker type and firm type.

The empirical challenge is to test for this interaction without making the evidence a saturated match-fixed-effect accounting exercise. A match intercept can absorb persistent pair-specific markdowns, but it mixes match surplus, firm pay policy, worker selection, and sparse within-match variation. We instead use the grouped distributional framework of Bonhomme, Lamadon, and Manresa (2019) as an auxiliary diagnostic. The BLM exercise keeps the outcome aligned with the rest of the paper—the log of the baseline allocated markdown, $\ln \mu_{ijt}^{\text{alloc}}$ —but asks whether grouped worker and firm heterogeneity enter additively.

The BLM sample keeps positive finite baseline markdown observations and uses the 2017–2019 pair, which gives the best balance between stayers and movers among two-year-apart candidate pairs. The grouped model uses ten firm classes, six worker types, and empirical distribution measures of firms’ 2017 log-markdown distributions. Appendix C.1 reports the sample, grouping diagnostics, and the canonical variance projection. The effective BLM sample contains 1.13 million log-markdown observations for the selected pair, including 913,980 stayers and 110,027 movers retained after grouping.

The BLM projection follows the variance decomposition in Bonhomme, Lamadon, and Manresa (2019): it linearly projects first-period outcomes simulated from the estimated grouped model—not the observed markdowns—on worker-type and origin-firm-class indicators. In our markdown application that projection gives a positive covariance term of 8.3 percent of log-markdown variance. This sign differs from the negative KSS-corrected AKM covariance in Section 4.1, but the two objects are not the same estimand. AKM uses individual worker and firm fixed effects in the full connected panel; BLM uses latent worker types, firm classes, and the model-implied composition of stayers and movers in a short two-year grouped model. Appendix Table C3 shows that the sign change is not generated by restricting the AKM effects to the BLM support; it appears when moving to the BLM simulated grouped projection.

Table 9 then asks whether the grouped model requires strong type-class interactions. This is the point at which the evidence speaks most directly to the concern raised by Bils, Kaymak, and Wu (2025): if markdowns are mainly organized by worker-firm comparative advantage, an additive worker-type plus firm-class representation should lose substantial explanatory power relative to a fully interacted type-class representation. It does not. Following the BLM paper’s comparison, the additive projection explains 53.7 percent of simulated log-markdown variance, while the fully interacted projection explains 55.3

percent. Interactions therefore raise R^2 by only 1.5 percentage points of total outcome variance. A separate diagnostic on the fitted worker-type-by-firm-class mean matrix shows why the interaction plot is still useful: conditional means are not perfectly additive, with an interaction residual equal to 38.0 percent of the variation in those fitted means. The two figures use different denominators: the 1.5-percentage-point figure is a share of total outcome variance, while the 38.0 percent figure is a share of the much smaller variation in the fitted type-class means. Non-additivity is thus a sizable share of a small object and a negligible share of the total.

TABLE 9. BLM additivity diagnostic, 2017–2019

Diagnostic	Estimate
A. Simulated BLM outcomes	
Additive worker-type and firm-class projection: R^2 (%)	53.7
Fully interacted type-class projection: R^2 (%)	55.3
Incremental R^2 from interactions (p.p.)	1.5
B. Fitted type-class mean matrix	
Additive share of fitted-matrix variation (%)	62.0
Interaction residual share of fitted-matrix variation (%)	38.0

Note: Panel A compares additive and fully interacted projections of simulated BLM outcomes, following the variance-decomposition exercise in Bonhomme, Lamadon, and Manresa (2019). Panel B applies the same additivity question to the fitted worker-type-by-firm-class mean log-markdown matrix. The fitted-matrix projection is weighted by the estimated type-class distribution; cells with numerically zero estimated mass are omitted, with retained mass rounding to 100.0 percent. Panel B entries are shares of fitted-matrix variation.

The BLM results therefore clarify how far the additive evidence can go. Additive firm and worker components summarize important persistent variation, and Table 9 shows that interactions add little explanatory power for variance accounting in this application. The positive covariance in the canonical BLM projection should not be read as overturning the negative AKM sorting result: it is a model-implied mobility-composition object, while the fitted type-class mean table and the AKM support checks do not show the same positive covariance. The evidence therefore does not reject the comparative-advantage logic in Bils, Kaymak, and Wu (2025); it says that, for these Portuguese worker-level markdowns and this grouped BLM design, type-class interactions add little explanatory power beyond additive worker-type and firm-class components.

6. Conclusion

Labor-market power has an incidence problem. Firm-level markdowns tell us whether an employer pays labor less than an output-based benchmark, but they do not say which workers inside the employer bear the wedge. This paper builds a worker-level markdown to answer that question. The measure allocates the part of firm value added available to

labor across occupations and workers, adds back to the firm-year labor claim, and has a marginal-product interpretation under explicit benchmark assumptions.

The evidence changes the picture delivered by a single firm wedge. Worker-level markdown dispersion has large firm and worker components, but the covariance between them is negative. Higher-occupation workers have larger wage-product wedges even after the allocation gives occupation groups different production weights. Market concentration and firm size also matter, but their gradients differ across occupations and within local markets. These patterns are hard to see in a firm average because the average mixes workers with different tasks, claims on value added, and wages.

The non-additive evidence sharpens rather than overturns this message. The canonical BLM simulated projection yields a positive covariance between worker types and origin firm classes, while the KSS-corrected AKM decomposition gives a negative firm-worker covariance. Appendix diagnostics show that this difference comes from the BLM grouped projection, not from applying the AKM components to the BLM support. Sorting is therefore not a primitive fact independent of the model used to summarize the data. But interactions add little explanatory power relative to additive worker-type and firm-class components. For these Portuguese markdowns, additive separability is a useful variance-accounting approximation, even though comparative advantage is the right economic force to check.

The main implication is methodological and substantive. Methodologically, production-side markdown measurement can be brought to the worker-firm panel without losing accounting discipline. Substantively, monopsony is not only a property of firms or markets. It is distributed across workers inside firms. Measuring that distribution makes it possible to ask whether labor-market power falls on low-wage workers, high-skill workers, occupations, or specific worker-firm matches. That is the object future empirical work on monopsony needs when the question is not only how much power firms have, but who pays for it.

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Appendix A. Proofs

For compactness, write

$$X_{jt} \equiv \prod_{g \in \mathcal{G}_{jt}} L_{gjt}^{\bar{\beta}_{g,s(j)}}.$$

A.1. Proof of Proposition 1

Under Assumption 1 and Assumptions 2 and 3,

$$VA_{jt}^L = A_{jt} X_{jt}, \quad X_{jt} = \prod_{g \in \mathcal{G}_{jt}} L_{gjt}^{\bar{\beta}_{g,s(j)}}.$$

Taking the derivative with respect to worker i 's hours gives

$$\frac{\partial VA_{jt}^L}{\partial h_{ijt}} = A_{jt} \frac{\partial X_{jt}}{\partial L_{g(i)jt}} \frac{\partial L_{g(i)jt}}{\partial h_{ijt}}.$$

Because $\partial L_{g(i)jt} / \partial h_{ijt} = q_{it}$ and

$$\frac{\partial X_{jt}}{\partial L_{g(i)jt}} = X_{jt} \bar{\beta}_{g(i),s(j)} \frac{1}{L_{g(i)jt}},$$

we obtain

$$\frac{\partial VA_{jt}^L}{\partial h_{ijt}} = A_{jt} X_{jt} \bar{\beta}_{g(i),s(j)} \frac{1}{L_{g(i)jt}} q_{it}.$$

Since $A_{jt} X_{jt} = VA_{jt}^L$ and, at observed hours,

$$L_{g(i)jt} = \sum_{k \in g(i),j,t} q_{kt} h_{kjt},$$

this becomes

$$\frac{\partial VA_{jt}^L}{\partial h_{ijt}} = VA_{jt}^L \cdot \bar{\beta}_{g(i),s(j)} \cdot \frac{q_{it}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}} = MRPL_{ijt}^{h,\text{true}}.$$

Multiplying by observed hours gives

$$h_{ijt} MRPL_{ijt}^{h,\text{true}} = VA_{jt}^L \cdot \bar{\beta}_{g(i),s(j)} \cdot \frac{q_{it} h_{ijt}}{\sum_{k \in g(i),j,t} q_{kt} h_{kjt}} = MRPL_{ijt}^{\text{alloc}}.$$

Finally,

$$\sum_{i \in j, t} MRPL_{ijt}^{\text{alloc}} = VA_{jt}^L \sum_{g \in \mathcal{G}_{jt}} \bar{\beta}_{g, s(j)} \sum_{i \in g, j, t} \frac{q_{it} h_{ijt}}{\sum_{k \in g, j, t} q_{kt} h_{kjt}} = VA_{jt}^L.$$

The first equality in the last display uses the within-group shares summing to one, and the second uses $\sum_g \bar{\beta}_{g, s(j)} = 1$. This proves the proposition.

A.2. Proof of Proposition 2

Under Assumption 1 and Assumptions 3 and 4,

$$VA_{jt}^L = \tilde{A}_{jt} \mathcal{H}(X_{jt}).$$

Differentiating with respect to worker i 's hours yields

$$\frac{\partial VA_{jt}^L}{\partial h_{ijt}} = \tilde{A}_{jt} \mathcal{H}'(X_{jt}) \frac{\partial X_{jt}}{\partial h_{ijt}}.$$

For $g = g(i)$, Assumption 3 gives $\partial L_{gjt} / \partial h_{ijt} = q_{it}$. Differentiating the inner Cobb-Douglas index gives

$$\frac{\partial X_{jt}}{\partial L_{gjt}} = X_{jt} \bar{\beta}_{g, s(j)} \frac{1}{L_{gjt}}.$$

Thus

$$\frac{\partial X_{jt}}{\partial h_{ijt}} = X_{jt} \cdot \bar{\beta}_{g(i), s(j)} \cdot \frac{q_{it}}{\sum_{k \in g(i), j, t} q_{kt} h_{kjt}}.$$

Therefore,

$$MRPL_{ijt}^{h, \text{true}} = \tilde{A}_{jt} \mathcal{H}'(X_{jt}) X_{jt} \cdot \bar{\beta}_{g(i), s(j)} \cdot \frac{q_{it}}{\sum_{k \in g(i), j, t} q_{kt} h_{kjt}}.$$

Using

$$VA_{jt}^L = \tilde{A}_{jt} \mathcal{H}(X_{jt}),$$

the allocated hourly measure satisfies

$$MRPL_{ijt}^{h, \text{alloc}} = \frac{MRPL_{ijt}^{\text{alloc}}}{h_{ijt}} = VA_{jt}^L \cdot \bar{\beta}_{g(i), s(j)} \cdot \frac{q_{it}}{\sum_{k \in g(i), j, t} q_{kt} h_{kjt}}.$$

Taking ratios gives

$$\frac{MRPL_{ijt}^{h, \text{true}}}{MRPL_{ijt}^{h, \text{alloc}}} = \frac{\mathcal{H}'(X_{jt}) X_{jt}}{\mathcal{H}(X_{jt})} \equiv \xi_{jt},$$

which is the elasticity of $\mathcal{H}$ at X_{jt} and is common within firm-year. Hence,

$$MRPL_{ijt}^{h,\text{true}} = \xi_{jt} MRPL_{ijt}^{h,\text{alloc}}, \quad h_{ijt} MRPL_{ijt}^{h,\text{true}} = \xi_{jt} MRPL_{ijt}^{\text{alloc}}.$$

The same common scalar multiplies every worker in the firm-year, so ratios of true hourly marginal products and ratios of true annual claims equal the corresponding allocated ratios. This proves the proposition.

Separability-scaled sensitivity measure. Proposition 2 also motivates a scaled sensitivity measure,

$$(A1) \quad \mu_{ijt}^{\text{sep}} \equiv \tilde{\xi}_{jt} \mu_{ijt}^{\text{alloc}},$$

where $\tilde{\xi}_{jt}$ is an analyst-supplied firm-year scaling proxy. This rescaling can be useful for sensitivity analysis under separability, but it should not be described as structural unless the relevant firm-year scalar is identified rather than proxied.

A.3. Proof of Proposition 3

Take the derivative of the firm objective in (10) with respect to W_{ijt} . Because the benchmark rules out cross-earnings effects across workers, only worker i 's term contributes to the derivative:

$$\frac{\partial}{\partial W_{ijt}} \sum_{k \in j, t} (MRPL_{kjt}^{\text{true}} - W_{kjt}) \ell_{kjt}(W_{kjt}; z_{kjt}) = -\ell_{ijt}(W_{ijt}; z_{ijt}) + (MRPL_{ijt}^{\text{true}} - W_{ijt}) \frac{\partial \ell_{ijt}}{\partial W_{ijt}}.$$

At an interior optimum this derivative equals zero, so

$$\ell_{ijt} = (MRPL_{ijt}^{\text{true}} - W_{ijt}) \frac{\partial \ell_{ijt}}{\partial W_{ijt}}.$$

Rearranging gives

$$MRPL_{ijt}^{\text{true}} = W_{ijt} + \frac{\ell_{ijt}}{\partial \ell_{ijt} / \partial W_{ijt}} = W_{ijt} \left(1 + \frac{1}{\eta_{ijt}} \right),$$

where

$$\eta_{ijt} \equiv \frac{\partial \ell_{ijt}}{\partial W_{ijt}} \frac{W_{ijt}}{\ell_{ijt}}.$$

The maintained positivity conditions imply that η_{ijt} is finite and positive, so the divisions are well-defined. Dividing both sides by W_{ijt} yields

$$\mu_{ijt}^{\text{true}} \equiv \frac{MRPL_{ijt}^{\text{true}}}{W_{ijt}} = 1 + \frac{1}{\eta_{ijt}}.$$

This proves the proposition.

Appendix B. Data appendix

TABLE B1. Sample construction

	Observations	Workers	Firms	Years
A. Sample construction				
1. QP worker-year raw	71,208,486	9,109,810	1,047,005	1985–2024
2. Worker-year with individual quality	7,834,266	3,761,370	381,727	1986–2024
3. Match with cleaned SCIE firm-years	40,296,974	5,647,907	441,119	2004–2024
4. Observation with valid MRPL	28,399,585	4,379,857	264,421	2004–2024
5. Connected set	13,724,903	2,367,228	166,035	2004–2024

Note: Rows report worker-year observations, unique workers, unique firms, and sample years. The rows are not all nested: the individual-quality row documents the availability of the entry-wage quality diagnostic, while the baseline markdown uses calibrated human-capital-adjusted hours. The final row is the connected worker-firm set used in the main additive decompositions.

Table B2 reports the raw-hours sensitivity for the markdown distribution in Table 2. The alternative keeps the same labor-claimable value-added numerator and occupation production weights, but allocates each firm-year-occupation claim in proportion to raw hours rather than human-capital-adjusted hours.

TABLE B2. Raw-hours sensitivity for estimated individual markdowns

Sensitivity to raw-hours allocation (log units)						
Occupation group	Mean baseline	Mean alternative	Δ mean	Median baseline	Median alternative	Δ median
Service/manual workers	-0.02	-0.02	0.00	-0.05	-0.06	-0.01
Technicians/clerks	0.57	0.58	0.01	0.56	0.58	0.02
Managers/professionals	0.70	0.70	0.00	0.66	0.67	0.01

Note: The baseline uses human-capital-adjusted hours inside firm-year-occupation cells. The alternative uses raw hours in the same cells. Entries are log-markdown means and medians by occupation group; differences are alternative minus baseline in log points.

Appendix C. Additional diagnostics

C.1. BLM diagnostics

Table C1 reports the sample behind the BLM exercise. The input panel contains 21.0 million positive-markdown observations for 3.6 million workers and 230,942 firms. The year-pair scan selects 2017–2019 because, among pairs no more than two years apart, it gives the largest minimum of stayers and movers. The selected pair has 1,069,718 workers observed in both years, including 913,980 stayers and 155,738 movers before grouping. The next-best pairs, 2016–2018 and 2018–2020, have 143,123 and 127,502 movers. After firm grouping, the effective mover count is 110,027.

TABLE C1. BLM sample for the non-additive markdown decomposition, 2017–2019

Statistic	Value
Input markdown observations	20,991,742
Unique workers	3,636,609
Unique firms	230,942
Selected years	2017 and 2019
Workers observed in both years	1,069,718
Stayers	913,980
Movers	155,738
Effective movers after grouping	110,027
Effective outcome observations	1,134,034
Outcome variance	0.729

Note: The BLM exercise filters the baseline markdown to positive finite values and uses $\ln \mu_{ijt}^{\text{alloc}}$ for the selected 2017–2019 pair. Effective movers are movers retained after firm grouping; effective outcome observations are the log-markdown observations used in the grouped BLM variance calculation.

The firm grouping step is important because the firm-level precluster data are thin for many firms. In the 2017 precluster year, the BLM input contains 1.48 million observations across 114,343 firms. About 56.0 percent of log-markdown variance is between firms and 44.0 percent is within firms, but the median firm has only four observations and 78.7 percent of firms have fewer than ten observations. Grouping firms into classes stabilizes the non-additive diagnostic while preserving firm-side heterogeneity.

C.2. KSS corrections

Table C4 reports leave-out diagnostics for the connected-set AKM decomposition. The KSS correction mainly lowers the worker variance share and makes the firm-worker covariance less negative, while leaving the conclusion that both firm and worker components are large.

Table C5 reports the same correction after adding the controls used in the occupation

TABLE C2. BLM variance projection, 2017–2019

Component	Variance	Share of total variance (%)
Worker type	0.052	7.2
Firm class	0.279	38.3
Sorting	0.060	8.3
Residual	0.337	46.3
Total variance	0.729	100.0

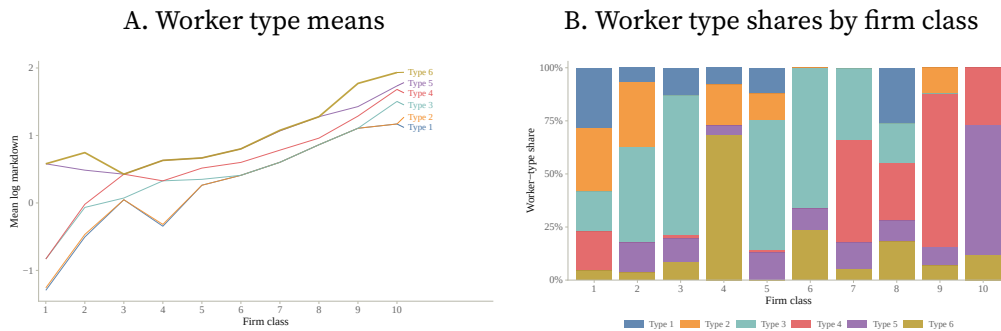
Note: BLM estimates use ten firm classes and six worker types for the selected 2017–2019 pair. Shares use the effective grouped BLM sample variance of log markdowns as the denominator. Following Bonhomme, Lamadon, and Manresa (2019), the table simulates first-period outcomes from the grouped model and projects them on worker-type and origin-firm-class indicators. Sorting is twice the covariance between the projected worker-type and origin-firm-class terms.

TABLE C3. BLM sorting bridge, 2017–2019

Object	Sorting share (%)	Interpretation
Full AKM connected sample	-34.8	Baseline additive AKM decomposition.
AKM effects on 2017–2019 connected support	-35.6	Same AKM effects, restricted to the BLM years.
AKM effects on BLM effective grouped support	-35.7	Same AKM effects, restricted to the grouped BLM support.
BLM simulated grouped projection	+8.3	Canonical BLM variance projection of simulated outcomes.
BLM fitted type-class mean matrix	-3.4	Additive projection of the fitted type-class mean table.

Note: The AKM rows evaluate the full-panel AKM worker and firm effects on progressively narrower samples. The BLM simulated grouped projection is the canonical BLM variance projection reported in Table C2. The fitted type-class mean matrix row applies the same additive covariance question to the estimated worker-type-by-firm-class mean log-markdown table.

FIGURE C1. BLM worker types and firm classes, 2017–2019



Note: Panel A plots the fitted worker-type-by-firm-class mean log-markdown matrix, with firm classes on the horizontal axis and separate lines for worker types. Panel B plots estimated worker-type shares within each firm class.

TABLE C4. KSS diagnostics for the AKM decomposition

A. Connected-set size and leverage diagnostics			
Sample	Observations	Workers	Firms
Baseline	3,082,673	1,832,924	116,144
Alternative	3,092,933	1,837,336	116,464
Sample	Mean P_{ii}	Median P_{ii}	Share $P_{ii} = 1$ (%)
Baseline	0.63	0.63	32.5
Alternative	0.63	0.62	32.5
B. Variance-share comparison: uncorrected vs KSS-corrected			
Sample-component	Uncorrected share (%)	KSS-corrected share (%)	Difference (pp)
Baseline: Firm effect variance	76.62	75.13	-1.49
Baseline: Worker effect variance	58.17	38.83	-19.35
Baseline: 2x firm-worker covariance	-34.80	-13.96	20.84
Alternative: Firm effect variance	80.47	80.07	-0.39
Alternative: Worker effect variance	55.60	34.80	-20.80
Alternative: 2x firm-worker covariance	-36.07	-14.87	21.19

Note: Panel A reports connected-set size and leave-out leverage moments from the KSS input. Panel B compares uncorrected and KSS-corrected AKM variance shares for the firm effect, worker effect, and twice the firm-worker covariance. The alternative sample uses the raw-hours allocation.

and local-market specifications. These entries should be read as corrected firm-worker components after partialling out the listed controls, not as full corrected decompositions of occupation or market effects. Across all three rows, the firm component remains the largest corrected component and the firm-worker covariance remains negative.

TABLE C5. KSS-corrected firm-worker shares after controls

Specification	Firm effect	Worker effect	2 × firm-worker covariance
Baseline AKM	75.13	38.83	-13.96
Occupation controls	75.02	41.91	-16.93
Local-market controls	78.81	35.86	-14.67

Note: Entries are variance shares in percent and can sum to 100 only after including the covariance term. The baseline row corrects the firm-worker AKM model. The occupation-controls row partials out tenure, occupation group, and year controls before correcting the firm-worker components. The local-market-controls row partials out tenure, occupation group, and industry × region × year controls. The table does not report a full corrected variance decomposition for occupation or local-market effects.

C.3. Fixed-effect distributions

Figure C2 and Table C6 report diagnostics for the estimated firm and worker effects. These diagnostics are useful for scale and dispersion; the effects are identified up to normalizations.

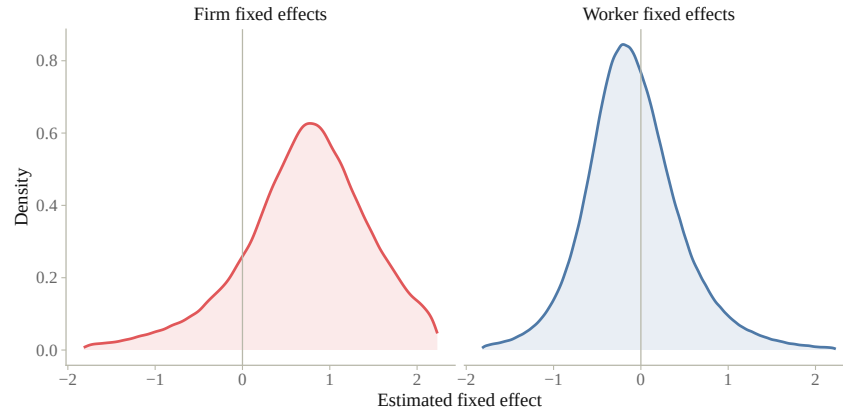


FIGURE C2. Distributions of estimated worker and firm fixed effects

Note: Kernel densities of worker effects $\hat{\chi}_i$ and firm effects $\hat{\alpha}_j$ from the baseline AKM decomposition. The figure trims the pooled 0.5th and 99.5th percentiles of estimated effects for readability.

TABLE C6. Occupation-specific moments of estimated worker and firm fixed effects

Occupation group	<i>N</i>	Mean	SD	P25	Median	P75
A. Worker fixed effects						
Service/manual workers	2,703,103	-0.19	0.58	-0.53	-0.24	0.12
Technicians/clerks	9,571,753	-0.10	0.52	-0.41	-0.13	0.17
Managers/professionals	1,450,047	-0.03	0.72	-0.49	-0.09	0.38
B. Firm fixed effects						
Service/manual workers	2,703,103	0.24	0.79	-0.35	0.35	0.79
Technicians/clerks	9,571,753	0.70	0.57	0.38	0.71	1.02
Managers/professionals	1,450,047	0.73	0.57	0.44	0.74	1.05

Note: Panel A reports moments of worker effects $\hat{\chi}_i$; Panel B reports moments of firm effects $\hat{\alpha}_j$. *N* counts worker-year observations in the connected set, so firm effects are observation-weighted. Occupation groups follow the three-category aggregation used in the main text.